



education
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GAUTENG PROVINCE

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ANNUAL TEACHING PLAN NATURAL SCIENCES GRADE 4- YEAR 1-2

2.1. Grade 4: Term 1

Strand: Life and living

Grade 4: Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
1	Living and non-living things	• Understand the field of Natural Sciences • List suggestions for classroom rules and repeat rules with prompting	The teacher shows images representing the topics in the four different strands to the learners to introduce them to the subject. The learners receive their homework books including letters to the parents to explain the nature of the subject.

Grade 4: Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		- Identify and name different kinds of living things - Identify and list the seven life processes – feeding, growing, reproducing, breathing, excreting, sensing and moving - Identify and list things that appear not to be living (e.g. dried beans, dried yeast, a fertilised bird egg), but carry on “living” given the right conditions - Identify and name different kinds of non-living things that cannot carry out all the seven life processes - Identify and name some things that were living and are now dead, e.g. dead wood, dry leaves	The teacher shows the learners real examples and pictures of living and non-living things and discusses the differences between living and non living things. <u>Grade 4: First year</u> <u>Activity 1</u> Group images and real life objects of living and non-living things into two groups. <u>Grade 4: Second year</u> <u>Activity 1</u> Discuss living and non-living things and list the differences and similarities between them.
2-3	Structures of plants, animals, birds and insects	- Identify and name the basic structure of plants: roots, stems, leaves, flowers, fruits and seeds - Identify and discuss the differences between plants, such as size, shape and colour of roots, stems, leaves, flowers, fruits and seeds - Identify the basic structure of animals, namely head, tail, body, limbs, sense	The teacher takes the learners to the school’s garden to identify the basic structures of plants and animals (e.g. chameleon, insects, grasshopper, lizards). <u>Grade 4: First year</u> <u>Activity 1</u> Identify and pull out a weed/plant in the garden and feel and smell the weed/plant. <u>Activity 2</u> Divide learners into small groups. Trace the outlines of the weed to make a poster. <u>Activity 3</u> Point out the basic structure of animals on images provided to them.

Grade 4: Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		organs - Identify the structure of birds, namely head, tail, wings, legs and sense organs - Identify the structure of insects, namely head, body, wings, legs and sense organs - Identify the visible differences between animals and birds, such as size, shape, body covering and sense organs	<u>Grade 4: Second year</u> <u>Activity 1</u> Identify and pull out a weed/plant in the garden, feel and smell the weed. Prepare and dry the weed/plant in a thick book. <u>Activity 2</u> Paste the weed in your workbooks and label each part of the weed. <u>Activity 3</u> Divide the learners in small groups and each group prepare and eat a mixed salad using different parts of the plant, e.g. lettuce leaves (leave of the plant), a tomato (fruit of the vegetable), celery (stem of the vegetable) and a carrot (root of the vegetable). <u>Activity 4</u> Discuss the differences and similarities between the animals shown on images provided by the teacher.
4	What plants, animals, birds and insects need to grow	- Identify and name what plants, animals, birds and insects need to grow, namely light, water and air - Identify and list what plants need to grow well, namely the right type of soil, right amount of water for the type of plant and sun or shade - Give examples how to grow plants, e.g. cuttings and seeds - Identify and list different plants that can	The teacher demonstrates why - plants need light, water and air to grow. Take four empty margarine containers and put seeds (e.g. beans, lentils, mealies) between two layers of damp cotton wool. Wait for 2 to 3 days for the seeds to sprout and remove the top layer of cotton wool. Place one container in a dark cupboard (no light) and three tubs near a window. Water all the tubs regularly, except one tub (no water). Put a plastic bag over one tub after the seeds sprouted to exclude air. The fourth tub serves as the control tub to show that plants need light, water and air to grow - plants need light to grow. Take an empty shoe boxes and cut a large, rectangular window in one short end of the shoe box. Take the piece of card that you have cut

Grade 4: Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		be used to take cuttings, e.g. cuttings from soft stems, runners, baby plants and leaves - List the factors seeds need to germinate, namely water and warmth - Know that plants make sugar and starch by using sunlight, water and carbon dioxide from the air - Know and understand the terminology: carbon dioxide, namely that carbon dioxide is the gas that we breathe out and the gas in fizzy drinks	out and cut a small window toward the end of the card. Secure the card with the window in the inside towards the middle of the long side of the box. Stand the box in the upright position. Place a small potted plant at the bottom of the box on the opposite side of the window card. The plant will grow in the shape of a half circle to reach the light. - animals, birds and insects need oxygen to breath. Take a small bottle, place three flies in the bottle and close the bottle tightly. Observe that the flies will die as they need oxygen to breath The teacher demonstrates how to record the results of the experiment, namely: - Which plants have grown - What the results of the experiment are <u>Grade 4: First year</u> <u>Activity 1</u> Divide learners into small groups and each group cut and grow a plant cutting taken from the school garden or home. Cuttings can be taken from geraniums, runners (e.g. creepers like an ivy or strawberry plant), spider plant and leaves from succulents. <u>Activity 2</u> Record how many days it takes for the cutting to develop a new leaf. <u>Grade 4: Second year</u> <u>Activity 1</u> Divide learners into small groups and each group cut and grow two cuttings from the same

Grade 4: Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			plant. The one cutting is put in a light, warm place and watered regularly. The second cutting is placed in a dark cupboard to establish the effect of light on the growth of plants. <u>Activity 2</u> Discuss and record how many days it takes for the cutting placed in an area with light to develop a new leave and how many days it takes for the other cutting in the cupboard to develop a leaf. <u>Activity 3</u> Divide learners into small groups and execute the experiment to test vegetables and fruit for sugar and starch. The teacher provides them with a slice of potato and apple and a few drops of iodine. Each group record their findings.
5	Habitats - plants, animals, birds and insects	• Understand and describe the word "habitat" • Identify and describe the habitat of plants • Describe habitats such as grassland, forest, river, desert and sea • List the differences between plants and weeds • List the plants that grow in your province • Know that the protea is our national flower • Identify and describe the habitat of animals • Identify and describe the habitat of birds • Identify the habitat of insects	<u>Grade 4: First year</u> <u>Activity 1</u> Provide learners with images of animals and their shelters. Match the animals to their shelters. <u>Activity 2</u> Identify at least one animal habitat on the school grounds and investigate the habitat in terms of materials, size, shape, colour and smell. <u>Grade 4: Second year</u> <u>Activity 1</u> Draw one animal habitat from images provided to them and describe the animal habitat. <u>Activity 2</u> Provide two reasons for the suitability of the animal habitat that you have chosen.

Grade 4: Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		- Discuss the different types of animal, bird and insect shelters (e.g. to shelter, to have babies and escape from danger) and differences in animal shelters - Know and understand the terminology habitat, indigenous plants and insects	
6	Structures for animal, bird and insect shelters	- Identify and list different types of natural animal, bird and insect shelters, namely nests, shells, hollow trees, wasp nests - Identify and name human made shelters, e.g. dog kennels, cages, kraals and stables - Identify and list the different shapes, sizes and materials animal structures can be made from	<u>Grade 4: First year</u> <u>Activity 1</u> Provide a small number of different materials to produce an animal, bird and insect shelter to the learners. Divide learners into small groups and each group make a simple model of an animal shelter, e.g. spider web, fish pond, bird nest. <u>Grade 4: Second year</u> <u>Activity 1</u> Provide the learners with images of different types of animal, bird and insect shelters. Describe two animal-, bird- and insect shelters. <u>Activity 2</u> List the steps to produce a model of one of the above animal, bird and insect shelters.
7-8	Healthy eating	Identify the food pyramid and the food in each group in the food pyramid, namely the bread and starch group, the fruit group, the vegetable group, the meat, chicken, fish and bean group, the dairy group and fats and sugar group	The teacher - provides the learners with different types of food to taste (sweet, sour and salty), smell (coffee, fresh leafy vegetable or onions, toast), see (previously mentioned products), hear (eating chips, drinking coffee) and feel (the texture of the mentioned products) on individual paper plates - places different food items in separate coloured plastic bags and the learners must

Grade 4: Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		- List the number of portions in each group to eat daily namely the bread and starch group (6-11 servings), the fruit group (2-4 portions), the vegetable group (3-5 groups), the meat, chicken, fish and bean group (2-3 servings), the dairy group (2-3 servings) and fats and sugar group (very little) - State tips for healthy eating daily - Identify the senses that we use when we eat food, namely the tongue to taste the food, nose to smell food	identify the food items by smelling and feeling. <u>Grade 4: First year</u> <u>Activity 1</u> Cut and group the pictures of the food items and paste in the correct position on the food pyramid. <u>Activity 2</u> Make a list with the food items that you ate yesterday and list the healthy food that you have not eaten. <u>Grade 4: Second year</u> <u>Activity 1</u> Construct a 3 dimensional food pyramid with the worksheet provided. Paste the pictures of the food items in the correct group. <u>Activity 2</u> Compile a list of the food that you ate yesterday. Compare your list with the food on the food pyramid to conclude if your food intake the previous day was healthy or not.
9	Recycling	- List litter that is dangerous to plants, animals and birds - Identify the recycling logo - Explain that recycling is a process to convert waste materials into reusable materials - List the meaning of the three arrows in the logo as “reduce, reuse and “recycle”	The teacher and learners - discuss how to collect litter - know how to behave during a cleaning-up expedition - collect litter from the school grounds to discuss how the litter may be dangerous to plants, animals and birds - group the litter in groups of paper, plastic, metal, glass and other <u>Grade 4: First year</u> <u>Activity 1</u>

Grade 4: Term 1			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		- List the groups of recyclable materials, namely cans, glass, paper, plastic, oil, electronic waste, drums and aerosols - Understand the terminology: litter, packaging, recycle	Draw up a list showing items that you have thrown in the rubbish bin yesterday and give a reason why these items may be harmful to plants, animals and birds. <u>Grade 4: Second year</u> <u>Activity 1</u> Draw up a list (with the headings paper, plastic, metal, glass and paper) showing all the items collected on the school grounds and group together. <u>Activity 2</u> Copy the recycling logo in your workbook and tell/write the meaning of the three arrows in the recycling logo.

2.2. Grade 4 Term 2

Strand: Matter and materials

Grade 4: Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
1 - 2	Materials around us	- Understand the terminology: matter and materials - Group matter as living and non-living matter - List examples of raw and manufactured materials - Know the phases of water as ice, water and steam - Know the properties of ice, namely it feels hard, cold and slippery and is white in colour - Understand what a solid, liquid and gas is - Give examples of solids, liquids and gases - List the characteristics of solids, liquids and gases: solids keep their shape, liquids flow and gases spread out and have no definite shape - List examples of materials that can change their shape, e.g. solids changes to a liquid when heated and changes to a	The teacher demonstrates - how to melt ice, by rubbing the ice between her/his hands, blowing on the ice or by placing the ice in the sun - the sensory properties of ice and that ice floats on water - how to boil water safely in a kettle and to pour water in a cup holding one hand behind his/her back. The teacher explains that the hand behind her back prevents her/him holder onto the cup and accidentally spilling boiling water on her/his hand. - that different types of liquids behave differently, e.g. syrup, oil and water by pouring the different liquids on a flat surface and observe how they flow differently <u>Grade 4: First year</u> <u>Activity 1</u> Group real life examples of solids and liquids (provided to learners) in two groups. <u>Activity 2</u> Divide learners in four groups and find images of gases in magazines to make a poster. <u>Activity 3</u> Boil water safely to make a cup of coffee and enjoy their coffee. <u>Grade 4: Second year</u> <u>Activity 1</u> Complete a worksheet and table the real life examples of solids and liquids (provided to learners).

Grade 4: Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		gas when cooled down. Gas changes to a liquid when cooled and the liquid changes to a solid when cooled further Know that the boiling point of water is 100°C and freezes at 0°C	<u>Activity 2</u> Divide learners in four groups. Find images of solids, liquids and gases in magazines to make a poster. <u>Activity 3</u> Boil water safely to make a cup of soup and enjoy their cup of soup. <u>Activity 4</u> Study an image depicting different matter (e.g. cars travelling on a road next to a river) and list the solids, liquids and gases in the picture.
3	Materials around us	- List the three states of water, namely water, ice and steam (water vapour) - List the solids, liquids and gases in the image of the water cycle - Know and understand the following terminology, condensation, evaporation and precipitation - Explain the water cycle	The teacher - boils water in a pot and leaves the water to cool down with the lid on the pot to explain condensation or blows against a mirror - put a little water in a saucer in a sunny corner in the class to demonstrate evaporation. <u>Grade 4: First year</u> <u>Activity 1</u> Put a weed in a small plastic bag and observe the water droplets forming in the bag. <u>Activity 2</u> Complete a worksheet to label each stage of the water cycle. <u>Activity 3</u> Demonstrate evaporation. <u>Grade 4: Second year</u> <u>Activity 1</u> Draw/copy and label an image of the water cycle in their books.

Grade 4: Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			<u>Activity 2</u> Plant a seedling in an empty coke bottle and close the open end with glad wrap to observe the water droplets forming against the glad wrap/condensation.
4-5	Materials around us	- Understand the terminology: raw and manufactured materials - Identify and list raw materials we use to make other useful materials, namely - sand is used to make glass - clay is used to make ceramics - coal and oil are used to make plastics, paints and fabrics - wood and fibre from plants are used to make paper - List different types of raw and manufactured materials - List different types of natural materials - Understand that combining materials results in new materials with added strength - Understand the terminology, namely processed materials – raw materials are processed to make other materials by using either heat or chemicals	The teacher - provides real life examples of raw and manufactured materials. The learners examine these examples and feel the texture of these materials. - shows a short video clip on how different materials are manufactured using raw materials, e.g. how fabric is manufactured, clay is used to manufacture clay pots, car tyres are manufactured and plant material is used to make homemade paper. <u>Grade 4: First year</u> <u>Activity 1</u> Divide learners in two groups. Group 1: Use mud only and shape the mud in the form of a brick. Group 2: Add straws to your bricks. Leave the bricks for a few days in the sun and then break the bricks. Compare the strength of each group's bricks. Understand that combining materials results in new materials with added strength. <u>Grade 4: Second year</u> <u>Activity 1</u> Add grass cuttings to water one week prior. Tear waste paper (preferable 80g A4 sheets) in small pieces, add the grass cuttings to the soaked waste paper and each learners spread 125 ml of the mixture on a flat absorbent surface (e.g. piece of cloth). Leave the "paper" to dry in the sun and remove the absorbent surface from the newly produced paper. Cut a shape from the paper and paste in your book.
6	Materials	Know that matter can cause fire, e.g.	The teacher demonstrates how to stop, drop and roll in case of clothes catching fire and the

Grade 4: Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
	around us	wood or paraffin - List the uses of wood and paraffin - List the danger of wood and paraffin, namely to cause a fire - Identify dangers in the home that may cause a fire, e.g. paraffin stoves, gas lamps and stove, loose electrical wires - Tell what to do in case of a fire in a house, namely to stop, drop and roll - Know fire safety messages, namely: -stay away from paraffin – it can hurt you -stay away from flames and fire – they can burn you -if clothes catch on fire, stop, drop and roll -cool a burn with cool water - Demonstrate how to use a fire extinguisher - Design an escape plan to exit the class safely in the case of a fire	learners practise to do the same. Learners are divided in small groups of four learners to complete the activities. <u>Grade 4: First year</u> <u>Activity 1</u> Identify and list dangers in the classroom/home that may cause a fire and mark these with a chalk. <u>Activity 2</u> Draw an escape plan to exit the class safely in the case of a fire. <u>Grade 4: Second year</u> <u>Activity 1</u> Describe how to adapt the dangerous objects in the classroom (that were identified by the younger group) to make them safe. <u>Activity 2</u> Draw an escape/evacuation plan to exit the school buildings safely in the case of a fire.
7	Solid materials	- Know and understand the meaning of the word: solid materials - Name the properties of raw and manufactured materials, e.g. hard or soft,	The teacher - demonstrates that heat changes the properties of materials by putting water in the freezer to become a solid that expands - demonstrates how to make sorbet using fruit juice to show that liquid expands when

Grade 4: Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		stiff or flexible, strong or weak, light or heavy, waterproof or absorbent - Demonstrate that heat causes matter to change, e.g. mealie meal versus cooked porridge, telephone lines contracting in cold weather and expanding in hot weather - Know that a sheep's wool is used to produce wool for jerseys/blankets and cotton from the cotton plant is used to make school shirts	frozen shows a number of bowls manufactured using different materials to the learners, e.g. a bowl made from metal, wood, wire, glass, plastic, clay and folded paper. The learners discuss the different properties of the raw materials used to produce the bowls and the uses of the bowls. <u>Grade 4: First year</u> <u>Activity 1</u> Make ice to demonstrate how heat changes the properties of water. <u>Activity 2</u> List the materials used to produce the bowls (displayed by the teacher) and name the properties of the bowls. <u>Grade 4: Second year</u> <u>Activity 1</u> Make sorbet to demonstrate how heat changes the properties of water. <u>Activity 2</u> Describe the properties of three raw and manufactured materials.
8	Strengthening materials	- Identify and list how to strengthen paper to build a strong structure - Roll paper in thin tubes (struts) to produce strong frames - Identify struts in images that are used to make strong, stable structures, e.g. roof trusses, bridges, cranes and pylons	The teacher demonstrates how to - fold and roll paper (one A4 sheet for each shape) in three different shapes, namely round, square and in a triangular shape. Let the sides overlap by 1 cm and place sticky tape on the top and the bottom. The teacher places a book on top of each pillar to determine the strength of the different shapes. The tube (round shape) is the strongest and will carry the most weight. - bend a straw in four equal lengths to make a triangle. The straw is placed on a table

Grade 4: Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		- Know that a square shape is strengthened by inserting a strut to form two triangular shapes - Know that a square is strengthened by inserting braces or gussets (small triangular shapes) in the corners of the square	and carefully pulled in the two opposite corners – the shape will change to a diamond shape. The teacher shows the learners how to strengthen the shape by inserting a straw diagonally into the square. <u>Grade 4: First year</u> <u>Activity 1</u> Work in pairs of two and repeat the teacher's experiment. Test the strength of the three different shapes by putting your lunch box on top of each pillar. Record in your book that the tube is the strongest of the three shapes. <u>Activity 2</u> Repeat the teachers experiment and paste the square with a diagonal strut in your workbook. <u>Grade 4: Second year</u> <u>Activity 1</u> Work in pairs of two. Repeat the teachers experiment, but use the round shape only to produce four thin tubes, using four A4 sheets. They place the four thin tubes between two lunch boxes, leaving a gap between the lunch boxes. Test the strength of the tubes by placing different weights on the tubes. Record your findings in your workbook. <u>Activity 2</u> Repeat the teachers experiment but instead of inserting a diagonal strut in the square, insert four small triangles in each corner of the square to strengthen the square.
9	Recycling	- Know that water is a scarce resource - Know that a drought causes damage or living things	The teacher demonstrates how to make water safe to drink by boiling the water for 15 minutes and then adding 5 ml bleach to 20 Litre water after the water has cooled down. Learners are divided in pairs to complete the activities.

Grade 4: Term 2			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		- Give one reason why polluted water cause ill health, namely that the water contains germs that causes cholera - List examples of how water pollution takes place - List ideas how to save water in the home and at school - List ideas how to save water in gardens and agriculture - Give examples of how to reuse water - Give examples of how to recycle water in the home - Know and understand the terminology: drought, water pollution, water recycling - List ideas how to use recycled cans and plastic - List suggestions how to re-use cans and plastic - Visit the website of the National Recycling Forum in South Africa at www.recycling.co.za to find your nearest recycler in your area	<u>Grade 4: First year</u> <u>Activity 1</u> Practise to make water safe to drink. Record the steps in your book. <u>Activity 2</u> Measure how much water a person uses to - flush the toilet once - wash four plates and cups Record these findings in your book. <u>Activity 3</u> Practise to group and recycle recyclables at school. <u>Grade 4: Second year</u> <u>Activity 1</u> Discuss suggestions how to safe water in the house and at school. Record these in your book. <u>Activity 2</u> Locate the water meter at school and take the reading of the water consumption per day over one week. Calculate how much water is consumed at school over one month. Record your findings in your book. <u>Activity 3</u> Practise to group and recycle recyclables at school.

2.3. Grade 4 term 3

Strand: Energy and change

Grade 4: Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
1	Energy and energy transfer	- Understand the terminology: energy and understand that energy makes things move - Know that humans need food for energy to grow, walk, run, sing and jump - Know that animals need food for energy to grow - Know that plants need food for energy to grow - Know that we need energy for everything we do, e.g. to cook food, switch on a light, open the tap to get hot water, fill our cars with petrol	The teacher demonstrates - activities that humans do that requires energy, e.g. walk, run, sing and jump - how energy moves by using a long piece of string tied between the backs of 2 chairs. Tie 2 balls to 2 shorter pieces of string. Tie these slightly apart onto the long piece of string that is tied between the 2 chairs. Start to swing the one weight gently. The second weight will soon start to swing without you touching it. Soon the first weight will stop swinging because all its energy has transferred to the second weight, after a while, the first weight will start to swing again. Energy from the second weight is transferred back to the first weight and the second weight stops swinging. <u>Grade 4: First year</u> <u>Activity 1</u> Identify activities that humans do that require energy and find, cut and paste pictures in their workbooks to illustrate energy. <u>Grade 4: Second year</u> <u>Activity 1</u> Identify activities that animals do that require energy and find, cut and paste pictures in their workbooks to illustrate this.
2	Energy and	Understand the concept: food chain	<u>Grade 4: First year</u>

Grade 4: Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
	energy transfer	- Identify the role players in a food chain, e.g. the sun shines to make plants grow that live stock feed on. We slaughter live stock to eat and obtain energy - Explain that plants make their own food by using energy from the sun, water from the soil and air - Explain pictures showing different food chains, e.g. how fish grow, how insects grow	<u>Activity 1</u> Cut out and place the pictures in a worksheet in sequence to represent the food chain. <u>Activity 2</u> Divide learners in groups of four. Place pictures of animals, birds, insects and plants in a box. Draw the pictures from a box and the groups takes turns to act out the picture they drew. <u>Grade 4: Second year</u> <u>Activity 1</u> Describe how plants make their own food. <u>Activity 2</u> Divide learners in groups of four to design a poster on how plants make their own food.
3	Energy around us	- Identify and name the different types of energy, namely heat, light, sound and movement energy - List the sources of heat energy and describe how these sources provide us with heat energy, e.g. sun, gas stove, paraffin stove, electric stove, fire - List the dangers of too much sun energy, namely heat stroke and skin cancer - List the sources of light energy and describe how these sources provide us with light energy e.g. sun, electric light,	<u>Grade 4: First year</u> <u>Activity 1</u> Identify and label pictures representing the different types of energy. <u>Activity 2</u> Make a list how to keep cool at school in very hot weather. <u>Grade 4: Second year</u> <u>Activity 1</u> Identify, label and describe pictures representing the different types of energy. The teacher provides the learners with key words to use to describe the pictures. <u>Activity 2</u> Make a list how to keep cool in your house during very hot weather.

Grade 4: Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		gas or paraffin lamps - List the sources of sound energy and describe how these sources provide us with sound energy e.g. musical instruments, radios, televisions, car sounds - List the sources of movement energy and describe how these sources provide us with movement energy e.g. petrol provides energy to buses and taxis, food provides humans with energy to push and pull objects	
4	Energy around us	- Understand the terminology: input and output energy - Identify and list sources of input energy, e.g. batteries in a phone, electricity to boil the kettle - Identify and list examples of output energy, e.g. send message with a phone, boiled water is the output energy	<u>Grade 4: First year</u> <u>Activity 1</u> Label pictures as either input or output energy. <u>Activity 2</u> Work in pairs of two learners. Make a model to illustrate input and output energy. <u>Grade 4: Second year</u> <u>Activity 1</u> Explain the differences between input and output energy. <u>Activity 2</u> Draw a table with two columns and list these words in either the input or output energy column: bicycle moves forward, leg movement to cycle a bicycle, batteries to operate a

Grade 4: Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
			torch, torch shines a light, electricity to operate a kettle, water boils in a kettle, flames heat the surrounding area, wood for a fire
5	Movement and energy	- Understand that musical instruments use movement energy such as blowing, beating and plucking to make sound - List five musical instruments and the type of movement energy needed, e.g. guitar needs plucking, drum needs beating, flute needs blowing - Identify sound as the main output energy of musical instruments - Know that different shapes and sizes produce different sounds - Produce musical instruments making different sounds	The teacher provides the learners with mealie pips, gravel and elastic bands. The learners produce their own musical instruments which are kept in the classroom to continue with sound identification the following week. <u>Grade 4: First year</u> <u>Activity 1</u> Bring two containers with lids from home with different shapes (e.g. square and flat, small round tin, tall and thin) to make your own musical instrument. Teacher provides mealie pips or popcorn pips to put in the containers. The learners shake their musical instruments to produce music and listen to the different sounds as output energy that is produced. <u>Grade 4: Second year</u> <u>Activity 1</u> Bring one containers without a lid from home to make your own musical instrument. Teacher provides different thicknesses of elastic bands to pull over the containers. Learners pluck the elastic bands to produce sound energy. <u>Activity 2</u> Draw your musical instruments and describe the shape of the container and the thickness of the elastic bands.
6	Energy and sound	- Explain that musical instruments make sound through vibration - Feel and hear the vibrations musical instruments make	The teacher demonstrates how to make a telephone with two - yoghurt containers and string - bigger containers and string Learners play with the telephone after they have completed their activities

Grade 4: Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		- Identify loud and soft sounds in and around the classroom - Identify high and low pitch sounds in and around the classroom - Identify different mediums sound travels through, e.g. air, water, plastic, metal and wood - Experiment with different types of sound to recognise the sounds	<u>Grade 4: First year</u> <u>Activity 1</u> Play on the musical instruments that you produced the previous week and identify the sound as loud or soft, high or low pitched. Record your findings in your book. <u>Activity 2</u> Place your musical instruments in a central place, and produce sounds with the musical instrument behind a screen. The other learners must determine the shape and the size of the musical instrument. <u>Grade 4: Second year</u> <u>Activity 1</u> The teacher makes sounds behind a screen. Identify the objects used to make the sound as well as the medium the sound travel through, e.g. tap a teaspoon against a cup in a box, ringing a bell inside a tin and, alarm clock ticking inside a plastic bag. Record your answers in your book. The teacher thereafter shows the objects and mediums used to produce the sounds to the learners.
7	Energy and sound	- Identify how the volume of water and air in containers affects sound vibration - Recognise that volume affects how loud sounds are. - Recognise that quick vibrations produce high sounds and slow vibrations produce	The teacher - fill identical glass bottles with water to different levels and blow across the top edge of each bottle to produce sounds at different pitch levels - fill identical glass bottle with water and tap on each bottle to illustrate a bottle xylophone <u>Grade 4: First year</u>

Grade 4: Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		low sounds - Identify the sensory organ that is used to interpret sound - Label the different parts of the ear correctly - Match the functions of each part correctly to a drawing of the ear	<u>Activity 1</u> Identify which bottle produces the sound with the highest and lowest pitch. <u>Activity 2</u> Bring a narrow container from home and repeat the teacher's experiment to determine how the volume of liquid in the container affects the sound that is produced. Experiment by blowing harder and softer across the top edge of the container and record your findings in your book. <u>Grade 4: Second year</u> <u>Activity 1</u> Learners label the different parts of the human ear, namely outer part, ear canal and eardrum and list the function of each part. <u>Activity 2</u> Execute an experiment to show how the volume of air influences sound.
8	Energy and sound	- Describe noise pollution as sounds that are loud, unpleasant or harmful to our ears and continues for a long time - Identify noise pollution in the surrounding areas - List the effect of noise pollution on humans - Explain that noise pollution can cause permanent damage to hearing - Explain the terminology: law	<u>Grade 4: First year</u> <u>Activity 1</u> List three examples of noise pollution that you experience at home. <u>Activity 2</u> Give ideas of how to protect yourself from the noise pollution. <u>Grade 4: Second year</u> <u>Activity 1</u> Describe noise pollution in your own words. <u>Activity 2</u>

Grade 4: Term 3			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		State that laws protect people from noise pollution	Work in four groups and select pictures of aspects that can cause noise pollution. Make a poster.
9	Recycling	- Make a list with different types of electronic waste - List the health risks electronic waste poses - Identify electronic waste stations nearest to your home - Visit the electronic waste association website at www.ewasa.org for more information	<u>Grade 4: First year</u> <u>Activity 1</u> Discuss in groups how to collect electronic waste safely <u>Activity 2</u> Practise to group and recycle recyclables safely at school. <u>Grade 4: Second year</u> <u>Activity 1</u> Discuss in groups how to transport electronic waste safely. <u>Activity 2</u> Practise to group and recycle recyclables safely at school.

2.4. Grade 4 term 4

Strand: Earth and beyond

Grade 4: Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
1-3	Planet earth	- Name the shape of the earth, namely a round ball or sphere - Know and understand the terminology: planet, solar system, cross section, layers, inner core, outer core, earth surface/crust, sea level, globe and atmosphere - Know the meaning of the word “solar system” - Show the layers of the earth on a sketch - Explain that the inner core is very hot - Explain that the crust is the thinnest layer - Tell that most of the surface of the earth is covered with water (oceans and seas) - Tell that the surface of the sea is called the sea level - Know that the land we can see is made up of continents and islands - Identify and/or list seven continents on earth, namely Africa, Australia,	The teacher shows a model of the earth and a 3-minute video clip depicting the layers of the earth to the learners. <u>Grade 4: First year</u> <u>Activity 1</u> Show/identify visible features such as oceans, continents and islands on a globe. <u>Activity 2</u> Work in groups of 4 to produce 2 dimensional models of the earth. <u>Grade 4: Second year</u> <u>Activity 1</u> Complete a worksheet to label the different layers of the earth. <u>Activity 2</u> Work in groups of 4 to produce 3 dimensional models of the earth.

Grade 4: Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		Antarctica, Asia and Europe, North and South America • Explain the difference between continents and islands • State that a thin layer of air surrounds the earth • State that the earth has many different habitats for living things	
4	Planet earth	• Know and understand the terminology: space, universe and stars • Know and understand that the earth is a planet and that there are eight planets • State that these planets have their own moons • Know that earth is the only planet with life on the planet • State that the moon is the closest object to earth and smaller than earth • Tell that we can see the sun, moon and stars from the earth	The teacher shows a short video clip of the solar system to the learners. <u>Grade 4: First year</u> <u>Activity 1</u> Copy our solar system from a picture provided by the teacher. <u>Grade 4: Second year</u> <u>Activity 1</u> Draw our solar system from the three dimensional model provided to them as an example of the solar system.
5	The sun	• Know and understand the terminology: solar power • State that the sun is a star at the centre	The teacher demonstrates how to - estimate the time using the position of the sun using a plasticine ball (fist size), a piece of cardboard about 10cm X 10 cm and five straws. Put the plasticine ball on the piece of cardboard in a sunny area and insert the straw in the plasticine ball so

Grade 4: Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		of our solar system • Tell that the sun is made of hot gas and gives out heat and light • Know that the sun is far away, but the closest star to the earth • State the safety precautions required when looking at the sun • Tell that the sun rises in the east and sets in the west • State that all life needs the sun and cannot do without • Tell that the sun is used to indicate time and direction	that it has no shadow. Repeat to insert plastic straws every hour until the learners leave school. The straw standing upright indicates noon and the straw the lowest position indicates the earliest time in the morning. Record the times that you inserted the straws on the piece of cardboard. - make a solar panel. Fill a black baking tray with cold water one centimetre deep. Measure the temperature of the water (with your finger or a thermometer). Place a sheet of plastic over the tray and leave in the sun for an hour. Take the plastic off and measure the temperature again. This experiment illustrates how solar power works. <u>Grade 4: First year</u> <u>Activity 1</u> Repeat the teacher's demonstration in groups of 2 learners. <u>Grade 4: Second year</u> <u>Activity 1</u> Stick a stick in the ground and mark the shadow of the stick. Record the time. Check the shadow of the stick every hour and draw a line in the soil. Check the length of the shadow and the markings in the soil every day for one week and you will notice that the shadows will be the same length every day at the same time. <u>Activity 2</u> Make your own solar panel.
6	The earth and the sun	• Know and understand the terminology: orbit, rotate • Know that day and night on all the	The teacher shows a model of a sundial and demonstrates how to make a model. <u>Grade 4: First year</u> <u>Activity 1</u>

Grade 4: Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		continents are not at the same time - Show and describe how the earth moves around the sun - State that the earth takes 365 days to move around the sun once - Tell that the earth rotates while it is moving around the sun - Know and understand how we get day and night - Know and understand that summer and winter are not at the same time during the year on all the continents	Discuss how life on earth would be without the sun. <u>Activity 2</u> Draw a sundial and use your teacher's model as an example. <u>Grade 4: Second year</u> <u>Activity 1</u> Discuss the different forms of water on our planet. <u>Activity 2</u> Divide the class in small groups to make a model of a sundial.
7	The moon	- Know and understand the terminology: phases of the moon, new moon, full moon - State that the moon is - a ball of rock in space - has no air and water - is smaller than the earth - is closer to the earth than the sun - Understand and know that the sun shines on the moon - Explain why the moon seems to have different shapes as observed from the	<u>Grade 4: First year</u> <u>Activity 1</u> Label the stages of the moon on the image showing the moon orbiting the earth. <u>Activity 2</u> Draw a table with two rows and seven squares. Write the days of the week in the first row. Draw the shape of the moon each night over seven nights. <u>Grade 4: Second year</u> <u>Activity 1</u> Describe the terminology full moon and new moon. <u>Activity 2</u> Discuss moon legends in African cultures.

Grade 4: Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		earth Discuss moon legends in African cultures	<u>Activity 3</u> Observe and record the changing shape of light on the Moon each night for a month on a calendar.
8	Rocket systems	- Know and understand the terminology: telescope, rocket, launch - Explain the uses of a telescope - Explain the uses of a rocket	The teacher demonstrates how to make a rocket using balloons, drinking straws and fishing line. <u>Grade 4: First year</u> <u>Activity 1</u> Divide learners in groups of 4 to make models of rockets. All the learners use the same shape balloons. Measure and record how far the “rockets” travels along the fishing line. <u>Grade 4: Second year</u> <u>Activity 1</u> Divide learners in groups of 4 to make models of rockets. Learners use balloons with different sizes and shapes. Measure and record how far the “rockets” travels along the fishing line. <u>Activity 2</u> Discuss the reason why the rockets travelled different distances.
9	Recycling	- Make a list with different types of glass and paper that are suitable to recycle - List the health risks glass recycling poses - List the different types of paper suitable to recycle - Know the type of paper that is the most profitable to recycle	<u>Grade 4: First year</u> <u>Activity 1</u> Play “I spy with my little eye something beginning with” until all items made from glass in the classroom have been identified <u>Activity 2</u> Make a list with suggestions how to re-use glass bottles. <u>Activity 3</u>

Grade 4: Term 4			
Week	Topic	Content The learner must be able to:	Suggested activities, investigations practical work and demonstrations
		Discuss a plan to start recycling at your school	Identify the closest bottle bank to the school and your home. <u>Grade 4: Second year</u> <u>Activity 1</u> Play “I spy with my little eye something beginning with” until all items made from paper in the classroom have been identified <u>Activity 2</u> Make a list with suggestions how to re-use different types of paper. <u>Activity 3</u> Identify the closest paper recycling centre to the school and your home.